

EXPERIMENT 1

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Experiment 2 – Study the basic network command and Network configuration commands.

Aim

Study the basic network command and Network configuration commands like ping, variations of ip config, tracert, nslookup, netstat, arp, rarp, hostname, pathping and basic networking commands.

Objective

- 1. To understand the purpose and usage of basic network diagnostic and configuration commands in Windows 11.**
- 2. To study the working of networking commands such as Ping, IPConfig, Tracert, NSLookup, Netstat, ARP, Hostname, PathPing, Route, and Net Use.**
- 3. To learn how these commands are used for network configuration, connectivity testing, and troubleshooting.**
- 4. To interpret the output of various networking commands for analyzing and diagnosing network-related issues.**

Software

- Windows PowerShell
- Windows Operating System

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Practical / Experiment Steps

Step -1

Open Windows PowerShell

Step 2

Type Various Commands and analyse their output.

1 – ping command

A) ping www.google.com

```
PS C:\Users\Jagrath> ping -f 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=25ms TTL=118
Reply from 8.8.8.8: bytes=32 time=20ms TTL=118
Reply from 8.8.8.8: bytes=32 time=21ms TTL=118
Reply from 8.8.8.8: bytes=32 time=234ms TTL=118

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 234ms, Average = 75ms
```

B) ping -n 2 www.google.com

```
PS C:\Users\Jagrath> ping -l 64 8.8.8.8

Pinging 8.8.8.8 with 64 bytes of data:
Reply from 8.8.8.8: bytes=64 time=253ms TTL=118
Reply from 8.8.8.8: bytes=64 time=233ms TTL=118
Reply from 8.8.8.8: bytes=64 time=22ms TTL=118
Reply from 8.8.8.8: bytes=64 time=21ms TTL=118

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 21ms, Maximum = 253ms, Average = 132ms
```

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C) ping -l 64 8.8.8.8

```
PS C:\Users\Jagrath> ping -n 2 www.google.com

Pinging www.google.com [142.251.152.119] with 32 bytes of data:
Reply from 142.251.152.119: bytes=32 time=282ms TTL=119
Reply from 142.251.152.119: bytes=32 time=27ms TTL=119

Ping statistics for 142.251.152.119:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 27ms, Maximum = 282ms, Average = 154ms
```

D) ping -f 8.8.8.8

```
PS C:\Users\Jagrath> ping www.google.com

Pinging www.google.com [142.251.152.119] with 32 bytes of data:
Reply from 142.251.152.119: bytes=32 time=242ms TTL=119
Reply from 142.251.152.119: bytes=32 time=21ms TTL=119
Reply from 142.251.152.119: bytes=32 time=233ms TTL=119
Reply from 142.251.152.119: bytes=32 time=64ms TTL=119

Ping statistics for 142.251.152.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 21ms, Maximum = 242ms, Average = 140ms
```

2 – ipconfig

A) ipconfig

```
PS C:\Users\Jagrath> ipconfig /all

Windows IP Configuration

Host Name . . . . . : Jaguuu
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Ethernet Connection (16) I219-V
```

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B) ipconfig /all

```
PS C:\Users\Jagrath> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:
```

C) ipconfig /release

D) ipconfig /renew

-Note: commands C) and D) release and get the new IP address respectively and may output a error if not all network media are connected, this error can be ignored.

3- tracert

A) tracert 8.8.8.8

```
PS C:\Users\Jagrath> tracert -h 2 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 2 hops:

  1    18 ms    5 ms    6 ms  10.18.159.16
  2   155 ms    6 ms    5 ms  dsldevice.lan [192.168.1.254]

Trace complete.
```


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B) tracert -h 8.8.8.8

```
PS C:\Users\Jagrath> tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  1    25 ms    14 ms     6 ms  10.18.159.16
  2    29 ms     8 ms     8 ms  dsldevice.lan [192.168.1.254]
  3    18 ms    18 ms    15 ms  100.94.0.1
  4    36 ms    23 ms    11 ms  192.168.201.60
  5     *        *        *    Request timed out.
  6    22 ms    21 ms   141 ms  192.178.83.35
  7    26 ms    32 ms    23 ms  142.251.52.213
  8   254 ms    21 ms    21 ms  dns.google [8.8.8.8]

Trace complete
```

C) tracert -d 8.8.8.8

```
PS C:\Users\Jagrath> tracert -d 8.8.8.8

Tracing route to 8.8.8.8 over a maximum of 30 hops

  1    15 ms     3 ms     2 ms  10.18.159.16
  2    23 ms     9 ms     7 ms  192.168.1.254
  3    24 ms    11 ms    10 ms  100.94.0.1
```

4- nslookup

A) nslookup www.google.com

```
PS C:\Users\Jagrath> nslookup www.google.com
Server:      Unknown
Address:     10.18.159.16

Non-authoritative answer:
Name:        www.google.com
Addresses:   2001:4860:482b:7700::
             2001:4860:482d:7700::
             2001:4860:482a:7700::
             2001:4860:4826:7700::
             2001:4860:4829:7700::
             2001:4860:4827:7700::
             2001:4860:4828:7700::
             2001:4860:482c:7700::
             142.251.157.119
             142.251.154.119
```

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B) nslookup 8.8.8.8

```
PS C:\Users\Jagrath> nslookup 8.8.8.8
Server:    UnKnown
Address:   10.18.159.16

Name:      dns.google
Address:   8.8.8.8
```

C) nslookup -type=A 8.8.8.8

```
PS C:\Users\Jagrath> nslookup -type=A 8.8.8.8
Server:    UnKnown
Address:   10.18.159.16

Name:      dns.google
Address:   8.8.8.8
```

D) nslookup -type=AAAA 8.8.8.8

```
PS C:\Users\Jagrath> nslookup -type=AAAA 8.8.8.8
Server:    UnKnown
Address:   10.18.159.16

Name:      dns.google
Address:   8.8.8.8
```

5- netstat

A) netstat

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```
PS C:\Users\Jagrath> netstat
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	10.18.159.98:49410	4.213.25.241:https	ESTABLISHED
TCP	10.18.159.98:52641	lb-140-82-114-26-iad:https	ESTABLISHED
TCP	10.18.159.98:55566	104.18.39.21:https	ESTABLISHED
TCP	10.18.159.98:56398	104.20.28.164:https	ESTABLISHED
TCP	10.18.159.98:58893	52.123.176.97:https	ESTABLISHED
TCP	10.18.159.98:58898	4.213.25.241:https	ESTABLISHED
TCP	10.18.159.98:59679	52.123.173.252:https	ESTABLISHED
TCP	10.18.159.98:60547	ec2-18-97-36-56:https	ESTABLISHED
TCP	10.18.159.98:62200	172.64.155.209:https	ESTABLISHED
TCP	10.18.159.98:64391	104.21.15.154:https	ESTABLISHED
TCP	127.0.0.1:49679	Jaguuu:49680	ESTABLISHED
TCP	127.0.0.1:49680	Jaguuu:49679	ESTABLISHED
TCP	127.0.0.1:49681	Jaguuu:49682	ESTABLISHED
TCP	127.0.0.1:49682	Jaguuu:49681	ESTABLISHED
TCP	127.0.0.1:49697	Jaguuu:49698	ESTABLISHED
TCP	127.0.0.1:49698	Jaguuu:49697	ESTABLISHED
TCP	127.0.0.1:49699	Jaguuu:49700	ESTABLISHED

B) netstat -n

```
PS C:\Users\Jagrath> netstat -n
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	10.18.159.98:49410	4.213.25.241:443	ESTABLISHED
TCP	10.18.159.98:55566	104.18.39.21:443	ESTABLISHED
TCP	10.18.159.98:55725	140.82.114.25:443	ESTABLISHED
TCP	10.18.159.98:56398	104.20.28.164:443	ESTABLISHED
TCP	10.18.159.98:58893	52.123.176.97:443	ESTABLISHED
TCP	10.18.159.98:58898	4.213.25.241:443	ESTABLISHED
TCP	10.18.159.98:59679	52.123.173.252:443	ESTABLISHED
TCP	10.18.159.98:63328	18.97.36.64:443	ESTABLISHED
TCP	10.18.159.98:64391	104.21.15.154:443	ESTABLISHED
TCP	127.0.0.1:49679	127.0.0.1:49680	ESTABLISHED
TCP	127.0.0.1:49680	127.0.0.1:49679	ESTABLISHED
TCP	127.0.0.1:49681	127.0.0.1:49682	ESTABLISHED
TCP	127.0.0.1:49682	127.0.0.1:49681	ESTABLISHED
TCP	127.0.0.1:49697	127.0.0.1:49698	ESTABLISHED
TCP	127.0.0.1:49698	127.0.0.1:49697	ESTABLISHED
TCP	127.0.0.1:49699	127.0.0.1:49700	ESTABLISHED
TCP	127.0.0.1:49700	127.0.0.1:49699	ESTABLISHED
TCP	127.0.0.1:49701	127.0.0.1:49702	ESTABLISHED

C) netstat -p TCP

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```
PS C:\Users\Jagrath> netstat -p TCP
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	10.18.159.98:49410	4.213.25.241:https	ESTABLISHED
TCP	10.18.159.98:50549	a23-55-106-105:https	ESTABLISHED
TCP	10.18.159.98:50550	57.155.101.212:https	TIME_WAIT
TCP	10.18.159.98:50552	a2-20-157-120:http	TIME_WAIT
TCP	10.18.159.98:50553	tzdelb-bk-in-f3:http	TIME_WAIT
TCP	10.18.159.98:50554	a23-55-106-98:http	TIME_WAIT
TCP	10.18.159.98:50555	199.232.214.172:http	TIME_WAIT
TCP	10.18.159.98:50556	20.190.146.36:https	ESTABLISHED
TCP	10.18.159.98:50557	20.190.146.36:https	ESTABLISHED
TCP	10.18.159.98:50558	57.155.101.212:https	TIME_WAIT
TCP	10.18.159.98:50559	57.155.101.212:https	TIME_WAIT
TCP	10.18.159.98:50561	57.155.101.212:https	TIME_WAIT
TCP	10.18.159.98:50562	57.155.101.212:https	TIME_WAIT
TCP	10.18.159.98:50566	server-54-192-142-12:https	TIME_WAIT

D) netstat -p UDP

6-arp

A) arp -a

```
PS C:\Users\Jagrath> arp -a
```

```
Interface: 10.18.159.98 --- 0xa
Internet Address      Physical Address      Type
10.18.159.16          56-f8-06-f2-24-db    dynamic
10.18.159.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

B) arp -d *

_Note: arp -d will delete the entries so don't run this command unless you REALLY know what you are doing.

7 – hostname

```
PS C:\Users\Jagrath> hostname
Jaguuu
```

8- pathping

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```
Computing statistics for 275 seconds...
Source to Here   This Node/Link
Hop  RTT      Lost/Sent = Pct  Lost/Sent = Pct  Address
0
1  25ms      0/ 100 = 0%      0/ 100 = 0%      |
2  68ms      1/ 100 = 1%      1/ 100 = 1%      dsldevice.lan [192.168.1.254]
3  44ms      2/ 100 = 2%      2/ 100 = 2%      100.94.0.1
4  45ms      0/ 100 = 0%      0/ 100 = 0%      192.168.201.60
5  41ms      0/ 100 = 0%      0/ 100 = 0%      192.168.200.1
6  39ms      2/ 100 = 2%      2/ 100 = 2%      121.23.41.103.netplus.co.in [103.41.23.121]
7  43ms      0/ 100 = 0%      0/ 100 = 0%      aes-static-125.34.144.59.airtel.in [59.144.34.125]
8  58ms      0/ 100 = 0%      0/ 100 = 0%      aes-static-125.34.144.59.airtel.in [59.144.34.125]
9  66ms      1/ 100 = 1%      1/ 100 = 1%      116.119.161.26
10 66ms      0/ 100 = 0%      0/ 100 = 0%      72.14.243.0
11 60ms      0/ 100 = 0%      0/ 100 = 0%      142.251.154.119
```

Learning Outcome

After completing this experiment, the learner will be able to:

1. Execute basic network configuration and diagnostic commands in a Windows 11 environment.
2. Interpret the output of commands such as **Ping**, **IPConfig**, **Tracert**, **NSLookup**, **Netstat**, **ARP**, **Hostname**, **PathPing**, **Route**, and **Net Use**.
3. Identify and troubleshoot basic network connectivity and configuration issues using command-line tools.
4. Explain the purpose and practical applications of commonly used networking commands.